

Quantifying Re-identification Risk in LLM-Generated Mobility Summaries through Iterative Questioning

Ayumu Kai
Oita University
Oita, Japan

ORCID: 0009-0001-3648-5842

Masanao Ochi
Oita University
Oita, Japan

ORCID: 0000-0002-6661-6735

Abstract—Mobility reports support transportation, urban development, retail, tourism, and public-policy decisions, and large language models (LLMs) can present complex mobility analyses to non-specialists. However, even without direct identifiers such as names and user IDs, analytical reports may still reveal information that could enable individual re-identification. This paper aims to quantitatively assess re-identification risk in LLM-generated mobility summaries. Quantifying this risk is difficult because free-form summaries contain different types and combinations of identifying cues. To this end, we propose an evaluation method that iteratively selects yes/no questions from a fixed question pool according to information gain. An LLM-based Oracle answers each question using only the released mobility summary, and the answers narrow the candidate set. We define $N(D)$ as the number of questions required to identify the target user, with a smaller value indicating higher re-identification risk among correctly resolved runs. Terminal-state diagnostics distinguish correct identification from false isolation and evidence exhaustion, with target elimination recorded as a separate, co-occurring diagnostic flag. We evaluated 100 target users against 200 candidates using structured mobility profiles under no perturbation and six levels of post-aggregation Laplace noise applied to aggregated count and frequency attributes, with a 200-question limit. Compared with no perturbation, the strongest tested perturbation ($\epsilon = 0.001$) reduced correct identification from 99% to 6%, increased false isolation from 1% to 82%, and increased resolved-only mean $N(D)$ from 11.63 to 17.83, with the latter based on six correctly resolved targets. These results demonstrate that the proposed evaluation method can quantitatively assess re-identification risk in LLM-generated mobility summaries. By mapping heterogeneous behavioral cues to $N(D)$ and diagnostic outcomes, the method contributes to quantifying the risk that persists after names and user IDs are removed. An implementation of the proposed audit pipeline is publicly available to facilitate reproducibility.

Index Terms—mobile location data, large language models, re-identification risk, numerical perturbation, information-gain evaluation

I. INTRODUCTION

The diffusion of smartphones and other networked devices has made large-scale GPS location data a routine part of contemporary digital infrastructure. Such data support decision-making in transportation, urban development, retail, tourism, and public policy [1], [2]. However, raw trajectories and mesh-level statistics remain difficult for non-specialists to

interpret because they require domain knowledge and careful contextualization [3]. LLMs can address this challenge by translating dense coordinate sequences and mesh-level records into human-readable descriptions of routines, destinations, and exceptional trips. Data-to-Text generation is therefore a promising way to make mobility data more accessible to analysts, policy makers, and citizens [4]–[7].

However, the same summaries that improve interpretability can also expose re-identification cues. A seemingly harmless description of weekday departure areas, weekend destinations, and transportation modes may allow an adversary to narrow a candidate set efficiently by combining contextual attributes [8]. The problem is not limited to explicit identifiers. Even when mesh codes or place names are removed, a summary may still reveal a distinctive combination of temporal, spatial, and behavioral features that functions as a quasi-identifier in narrative form.

The purpose of this paper is to quantitatively assess re-identification risk in LLM-generated mobility summaries.

The central challenge is to assess re-identification risk consistently across heterogeneous mobility summaries and privacy transformations while accounting for both successful and unsuccessful identification outcomes. We decompose this challenge into the following research questions:

- **RQ1:** How can re-identification risk in mobility summaries be quantified?
- **RQ2:** Does stronger numerical perturbation reduce re-identification risk in LLM-generated mobility summaries?
- **RQ3:** How do attribute removal and prompt abstraction change identification outcomes?
- **RQ4:** How do failure modes affect the interpretation of re-identification risk?

To address this challenge, we propose an evaluation method centered on $N(D)$, defined as the number of questions required to correctly isolate a target user. In this method, an evaluator iteratively selects yes/no questions from a fixed question pool according to information gain. An LLM-based Oracle answers each question using only the released summary, and the answers are used to narrow the candidate set. Among correctly resolved runs, a smaller $N(D)$ indicates higher re-

identification risk. By mapping heterogeneous identifying cues to a common question-based measure and reporting resolution and failure diagnostics, the method enables consistent assessment across mobility summaries and privacy transformations while accounting for both successful and unsuccessful identification outcomes.

We evaluated the method on GPS-derived structured mobility profiles for 100 target users against a fixed 200-user candidate pool, with a 200-question limit. The experiments covered six levels of inverse-epsilon-scaled post-aggregation Laplace perturbation of aggregated count and frequency attributes, attribute removal, and prompt abstraction. The method captured changes in identifiability: under the strongest tested perturbation ($\epsilon = 0.001$), relative to no perturbation, mean $N(D)$ among correctly resolved runs increased from 11.63 to 17.83, correct identification decreased from 99% to 6%, and false isolation increased from 1% to 82%. It also captured distinct changes in identification outcomes under attribute removal and prompt abstraction.

Overall, this work provides a quantitative indicator of the re-identification risk that can arise from the same mobility summaries that improve interpretability. This paper makes two contributions. First, we introduce $N(D)$ as a question-based measure that converts heterogeneous identifying cues into a common measure of identification effort. Second, we pair $N(D)$ with terminal-state diagnostics and empirically demonstrate the resulting framework across post-aggregation numerical perturbation of aggregated count and frequency attributes, attribute removal, and prompt abstraction.

An implementation of the proposed audit pipeline, including the prompts and configurations for the main numerical-perturbation experiment, is publicly available to facilitate reproducibility.¹

The remainder of this paper reviews background and related work, formulates $N(D)$, describes data preparation and the experimental design, reports the completed diagnostic results, and concludes with implications for pre-release privacy evaluation.

II. BACKGROUND AND RELATED WORK

This section reviews three foundations of our evaluation method: mobility re-identification and location protection, privacy in anonymized and generated text, and interactive identification through sequential questions.

A. Mobility Re-identification and Location Protection

Human mobility traces are highly distinctive: four spatiotemporal points uniquely characterized 95% of 1.5 million traces in one study [9]. This motivates assessing combinations of behavioral cues rather than direct identifiers alone.

DP formally bounds changes in an analysis output between adjacent datasets [10]; geo-indistinguishability adapts this principle to location release [11]. These mechanisms are important, but do not define the semantic identifiability of a

particular generated summary. Our post-aggregation Laplace perturbation of aggregated count and frequency attributes is therefore only an experimental baseline, not a DP or geo-indistinguishability guarantee.

B. Privacy in Anonymized and Generated Text

Entity removal is insufficient when quasi-identifiers can be combined through semantic inference [8]. TAB accordingly annotates direct and quasi-identifying spans and evaluates privacy and utility, not entity detection alone [12]. However, it does not measure how a whole free-form summary narrows a candidate population.

Privacy leakage also occurs in generated summaries. LLMs have retained PII in medical and legal summaries despite privacy-oriented prompting or fine-tuning [13]. Moreover, up to 80% of masked spans were recovered from de-identified documents using contextual text and retrieved background knowledge [14]. These studies establish the problem but do not quantify candidate-level risk from heterogeneous cues after names and user IDs are removed. Our method addresses this gap at the level of the released summary.

C. Interactive Identification by Sequential Questions

Akinator-like systems and Twenty Questions identify a hidden entity through repeated candidate-narrowing questions. De Bruyn et al. used Akinator traces to show that LLMs can answer such questions, while also exhibiting ambiguities and errors [15]. Generalized binary search formalizes the selection step by choosing queries that balance the remaining hypotheses and can attain logarithmic query complexity under suitable conditions [16]. This information-seeking view provides the basis for selecting questions by information gain.

Prior work treats question count as search efficiency. We repurpose it as privacy risk: fewer questions to correct isolation indicate higher risk, and we also record `unknown` answers and erroneous terminal states, which idealized search models do not represent as privacy outcomes.

III. PROPOSED EVALUATION METHOD: $N(D)$

Figure 1 shows the overall idea of the proposed evaluation method. The method starts from a structured mobility profile, optionally applies a protection mechanism to that profile, generates a natural-language summary, and then audits the released summary through an interactive narrowing process. The objective is not to evaluate the structured data or the protection mechanism in isolation, but to measure how difficult it remains to identify the target from the final text that would actually be released.

A. Audit Setting

A metric for free-text mobility summaries should be mechanism-agnostic, evaluate the released text itself, and define a reproducible search procedure. $N(D)$ therefore asks how many binary questions an information-gain-based auditor requires, after reading the summary, to isolate the target from the candidate population under a fixed question pool, Oracle

¹<https://github.com/m-ochilab/mobility-privacy-llm-a-kai>

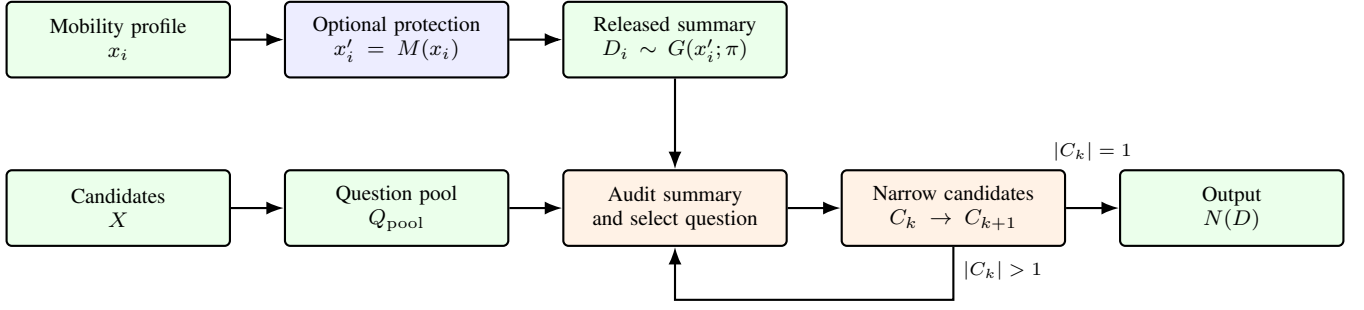


Fig. 1. Overall concept of the proposed $N(D)$ evaluation method. A structured mobility profile is optionally perturbed and converted into a released summary, which is then audited against a candidate pool through question-based narrowing. Non-resolved terminal states, including false isolation and zero information gain, are recorded as diagnostics. $N(D)$ is defined only for true target resolutions.

configuration, and answer protocol. The question-selection and candidate-update rules are deterministic given a fixed sequence of Oracle answers, though the end-to-end audit may still vary with the LLM-based Oracle’s behavior.

We adopt an explicit but restricted audit adversary. The auditor knows the candidate population and the automatically generated question pool, but does not know in advance which candidate corresponds to the released summary. At each step, the auditor selects the next question according to the information-gain criterion defined below, and answers it using only the released summary D as evidence. The answer space is restricted to `yes`, `no`, and `unknown`.

The adversary cannot use external knowledge such as social profiles, address books, workplace databases, or public records. Thus, the metric is not a worst-case attack bound; it is a controlled audit abstraction for comparing how many identifying cues are encoded in generated summaries.

B. Summary Generation

Rather than providing raw trajectories directly to the generator, we compress each user’s observations into a structured mobility profile. The profile covers location, time, movement, and frequency attributes, because re-identification often arises from the combination of where, when, how, and how often a person moves. The concrete extraction rules are described in Section IV.

Let $\mathcal{U} = \{u_1, \dots, u_n\}$ be a set of individuals, and let x_i denote the structured mobility profile of user u_i . Before generation, an optional protection mechanism M may transform the profile:

$$x'_i = M(x_i), \quad (1)$$

where M may represent numerical perturbation, attribute suppression, abstraction, or the identity mapping when no protection is applied. A summary generator G then produces the released text D_i from the protected profile x'_i under a generation policy π :

$$D_i \sim G(x'_i; \pi). \quad (2)$$

This formulation separates the protection stage from the text-generation stage, which is important because different privacy

controls may operate before or during generation while still being evaluated through the same output-side audit.

C. Audit Metric and Question Selection

Let $X = \{x_1, \dots, x_m\}$ denote the candidate population, let x^* denote the target profile, let Q_{pool} denote the fixed question pool, and let B denote the search budget. The audit begins with candidate set $C_0 = X$. At each step k , the auditor selects a question $q_k \in Q_k$, receives an answer $a_k \in \{\text{yes}, \text{no}, \text{unknown}\}$, and updates the candidate set. When the answer is informative, the update is

$$C_{k+1} = \{x \in C_k \mid \text{attr}(x, q_k) = a_k\}, \quad (3)$$

where $\text{attr}(x, q)$ denotes the answer implied by candidate profile x for question q . When the answer is `unknown`, the candidate set is left unchanged, that is, $C_{k+1} = C_k$.

The anonymity metric $N(D)$ is defined only for truly resolved runs. It is the number of audit steps required to correctly isolate the target under the fixed protocol:

$$N(D) = k \quad \text{if } C_k = \{x^*\} \text{ first occurs with } k \leq B. \quad (4)$$

Otherwise, $N(D)$ is undefined.

When the target is not correctly isolated within the fixed budget of $B = 200$, or when the sole remaining candidate is not the target, the run terminates with a diagnostic state and $N(D)$ remains undefined; we separately record the actual number of asked questions, including `unknown` answers that leave the candidate set unchanged. A run is counted as resolved only when the final candidate set contains exactly one profile and that profile is the target; a sole remaining candidate that is not the target is recorded as false isolation rather than success. Target elimination denotes a case in which the true target is removed from the candidate set during the audit. It is not an operational stopping condition, since the auditor does not know the target identity during the search; it is computed retrospectively and may co-occur with false isolation, zero-information-gain termination, or another terminal outcome. Disagreement between the Oracle answer and the target’s ground-truth feature value is counted in the field named `oracle_error_count` for diagnosis only and does not correct candidate updates. Because numerical perturbation

can intentionally make the summary reflect a perturbed rather than original profile, this field is better interpreted as a target-profile disagreement count than as a direct measure of Oracle accuracy. We therefore report Mean $N(D)$ only over truly resolved runs, always paired with the number of resolved runs, and separately record Oracle-unknown occurrence as a non-terminal diagnostic.

At each step, the auditor selects the question expected to reduce candidate uncertainty most efficiently. Let the uncertainty of candidate set C be represented by Shannon entropy:

$$H(C) = - \sum_i p(x_i) \log_2 p(x_i), \quad p(x_i) = \frac{1}{|C|}. \quad (5)$$

Here, $p(x_i) = 1/|C|$ assumes a uniform prior distribution over the active candidate set C . For a binary question q , let C_{yes} and C_{no} denote the candidate subsets consistent with the corresponding answers. The information gain (IG) of question q is defined as

$$IG(C, q) = H(C) - \sum_{v \in \{\text{yes}, \text{no}\}} \frac{|C_v|}{|C|} H(C_v), \quad (6)$$

$$q^* = \arg \max_{q \in Q_{\text{pool}}} IG(C, q). \quad (7)$$

Ties in maximum information gain were broken deterministically by ascending question-pool ID (QID) order.

IG is computed only for eligible questions that induce a complete binary partition of the current candidate set, i.e., every current candidate yields a defined `yes` or `no` answer, so that $|C| = |C_{\text{yes}}| + |C_{\text{no}}|$. Candidate-side missing values are excluded at question-selection time rather than treated as negative evidence, while Oracle-side `unknown` leaves the candidate set unchanged without affecting eligibility for the next step.

This rule makes the question-selection procedure adaptive and reproducible given fixed Oracle outputs: after each answer, the candidate set is updated and the information gain of the remaining questions is recomputed. The selection strategy is therefore fixed at the protocol level even though the resulting question sequence depends on the summary under audit.

The question pool Q_{pool} is automatically generated from the structured candidate profiles after converting location values to Google place names at the sublocality level. Numerical attributes yield threshold questions at the 0.25, 0.50, and 0.75 quantiles; categorical attributes yield equality questions; spatial questions use sublocality place names rather than raw mesh identifiers; and frequency questions cover visit counts, rare-event detection days, monthly record count, and active days. Ablation conditions use the same full question pool, so hidden attributes affect the released summary rather than the adversary’s available questions; if a hidden attribute lacks explicit textual evidence, the Oracle should return `unknown`.

During the evaluation, answers to questions are inferred from the generated summary D rather than directly copied from the structured feature table, so that $N(D)$ measures the identifiability of the released text itself rather than of the hidden database.

Algorithm 1 Re-identification Risk Audit for Computing $N(D)$

```

1: Input: Candidate profiles  $X = \{x_1, \dots, x_m\}$ , target  $x^*$ ,
   summary  $D$ , question pool  $Q_{\text{pool}}$ , fixed budget  $B$ 
2: Output: terminal state, questions asked, and  $N(D)$  if
   resolved
3:  $C \leftarrow X$ ,  $Q \leftarrow Q_{\text{pool}}$ ,  $k \leftarrow 0$ 
4: while  $|C| > 1$  and  $Q \neq \emptyset$  and  $k < B$  do
5:    $Q_{\text{eligible}} \leftarrow \{q \in Q \mid$ 
      $q \text{ induces a complete binary partition of } C\}$ 
6:   if  $Q_{\text{eligible}} = \emptyset$  then
7:     break
8:   end if
9:    $q^* \leftarrow \arg \max_{q \in Q_{\text{eligible}}} IG(C, q)$ , with ties resolved by
     the predefined ascending-QID question-pool order
10:  if  $IG(C, q^*) \leq 0$  then
11:    break
12:  end if
13:   $v \leftarrow \text{audit}(D, q^*)$ , where  $v \in \{\text{yes}, \text{no}, \text{unknown}\}$ 
14:  if  $v \neq \text{unknown}$  then
15:     $C \leftarrow \{x_i \in C \mid \text{attr}(x_i, q^*) = v\}$ 
16:  end if
17:   $Q \leftarrow Q \setminus \{q^*\}$ ;  $k \leftarrow k + 1$ 
18: end while
19: if  $C = \{x^*\}$  then
20:   return resolved, questions asked  $k$ , and  $N(D) \leftarrow k$ 
21: else
22:   return the applicable diagnostic terminal state, ques-
     tions asked  $k$ , and  $N(D) \leftarrow \text{undefined}$ 
23: end if

```

The Oracle answers `yes` only when the queried attribute value is explicitly stated in the released summary, and `no` only when the same attribute is explicitly stated with a different value. If the attribute is absent, hidden, abstract, ambiguous, or requires inference, the Oracle returns `unknown`; absence of evidence is not treated as `no`. For numerical threshold questions, only a direct comparison between an explicitly stated value and the threshold was allowed, excluding semantic, contextual, or commonsense inference. Unknown responses do not reduce the candidate set but still consume one audit step, reflecting an auditor constrained by a fixed query budget; they may therefore increase the number of steps in a resolved run or contribute to non-resolution. This stricter evidence rule reduces false isolation caused by unsupported yes/no answers, and Oracle fallibility is itself one reason the audit records terminal-state diagnostics.

D. Audit Algorithm

Algorithm 1 summarizes the procedure for computing $N(D)$. Starting with the full candidate set, the procedure repeatedly selects an eligible question, obtains an answer from the summary, and updates the candidate set.

The algorithm greedily selects the question with the largest information gain for the current candidate set. After a question

is used, it is removed from the pool, preventing repeated questions from artificially inflating the count. Given the same candidate set, question pool, summary, Oracle outputs, and tie-breaking rule, the resulting search trace is deterministic.

IV. DATASET AND PROFILE CONSTRUCTION

This section describes the common data preparation used in all experiments, including candidate selection and mobility-profile construction.

A. Data and Candidate Selection

We used mobile-phone GPS point data provided by Agoop.² The analysis period was one month, from November 1 to November 30, 2025, and we restricted the records to Android devices to reduce variation caused by OS-dependent positioning behavior and sampling policies. Each record contains an anonymized UUID, timestamp, latitude and longitude, fine-grained spatial mesh information, day-of-week information, and an estimated transportation-mode label. The raw records may include 100-m mesh information, but prompts, question pools, and summaries use only Google place names at the sublocality level. During preprocessing, records with missing UUIDs were removed, and only valid Android records within the target period were used. We defined a valid user as one with at least 50 GPS records during the month, sorted valid users by record count, and retained the top 200 as the candidate pool. The 100 target users were selected from this pool by seeded random sampling without replacement, avoiding further bias toward the highest-frequency users.

B. Mobility Profile Construction

For each user, we constructed a mobility profile from a sequence of representative longer-stay locations. We extracted third-level mesh areas (approximately 1 km) for weekday morning (6:00–9:00), weekday daytime (11:00–15:00), weekday night (19:00–23:00), and weekend daytime (11:00–15:00), using the most frequent mesh in each window as the representative location. We also retained the existing rare-event logic for a distant event outside the weekday routine. Before prompt rendering and question-pool generation, location values were converted into place-name labels at sublocality granularity using a cached reverse-geocoding pipeline. The reported target100 run used only the existing cache and issued no new API requests. Thus, summaries and audit questions used human-readable place names rather than raw mesh identifiers.

V. EXPERIMENTS

This section organizes the evaluation design around the four research questions. RQ1 uses a common audit protocol, RQ2 examines numerical perturbation, RQ3 examines attribute removal and prompt abstraction, and RQ4 analyzes failure modes.

A. Common Evaluation Protocol (RQ1)

All reported experiments used 100 targets, 200 candidates, and a 200-question limit. At each step, the highest-information-gain eligible question was selected and removed from the pool; an `unknown` answer left the candidate set unchanged. We used `google/gemini-3.1-flash-lite` via OpenRouter as both summary generator and summary-based Oracle. The Oracle answered from the released summary alone using `yes`, `no`, or `unknown`.

For every summary, the generator received the target user’s behavioral profile and a fixed role instruction. The prompt prohibited UUIDs, hallucinated place names, raw spatial identifiers, and attributes hidden by the condition, while allowing sublocality place names and visit counts when included in the prompt representation. The complete prompts, decoding parameters, parsing rules, retry policy, and configurations for the main numerical-perturbation experiment are provided in the public repository.³ The generator used temperature 0.2, while the Oracle used temperature 0.0 and a maximum of 32 completion tokens, with up to three retries for transient failures. An explicit decoding seed was requested from OpenRouter, but exact bit-level reproducibility is not guaranteed because the provider does not guarantee a fixed model snapshot.

B. Numerical Perturbation (RQ2)

The main quantitative evaluation used seven conditions: No Perturbation, $\epsilon = 100$, $\epsilon = 10$, $\epsilon = 1$, $\epsilon = 0.1$, $\epsilon = 0.01$, and $\epsilon = 0.001$. After constructing each user’s structured mobility profile, we applied post-aggregation Laplace noise to aggregated count- and frequency-based numeric attributes, including visit counts, rare-event detection days, monthly record count, and active days, with the scale inversely proportional to ϵ . Location- and traffic-related attributes were not directly perturbed. Generated summaries used profiles with perturbed count and frequency attributes, whereas candidate pools and target-side matching used the corresponding original profiles. This is a post-aggregation numerical-perturbation baseline, not a formal differential-privacy or geo-indistinguishability guarantee. The seven conditions produced 700 summaries and audit runs.

C. Attribute Removal and Prompt Abstraction (RQ3)

Experiment 2 reports four completed 100-target conditions: Full Baseline, w/o Location, w/o Time, and w/o Frequency. Location, time, and frequency attributes were hidden separately while retaining the same full question pool. We also report a preliminary 100-target prompt-granularity run (Levels 1–3) under the same candidate-pool size and question budget. The three levels use the same structured input features and differ only in requested output granularity. These runs are diagnostic evidence rather than conclusive rankings of privacy controls, and only completed conditions support quantitative claims. The Experiment 2 conditions came from separate

²<https://agoop.co.jp/service/dynamic-population-data/>

³<https://github.com/m-ochilab/mobility-privacy-llm-a-kai->

runs, so we report condition-wise outcomes without assuming complete run-level equivalence. Baselines from different experiments are not treated as interchangeable.

D. Failure-Mode Analysis (RQ4)

For RQ4, evaluation stopped at correct resolution, false isolation, an empty candidate set, zero information gain, question exhaustion, or the question limit. We report correct resolution, false isolation, target elimination, Mean Unknown Steps, and Mean $N(D)$ over correctly resolved runs. Identification effort is interpreted together with resolved counts and failure outcomes rather than by assigning unresolved runs an artificial question count.

VI. RESULTS AND DISCUSSION

This section presents the results for each research question. Each RQ subsection presents the observed outcomes, discusses their interpretation, and ends with a concise answer to the corresponding RQ. Terminal-state diagnostics are reported together with $N(D)$ throughout.

A. Quantifying Re-identification Risk with $N(D)$ (RQ1)

Table I reports the completed full-scale numerical-perturbation conditions. Post-aggregation Laplace noise was applied to aggregated count and frequency attributes; these conditions are experimental baselines and do not provide formal DP or geo-indistinguishability guarantees.

Across the seven Experiment 1 conditions, $N(D)$ was obtained only for correctly resolved runs. Its mean ranged from 11.54 to 17.83 across the seven conditions, while unresolved runs were represented by terminal-state diagnostics rather than an artificial question count. The prompt-granularity and attribute-ablation results reported below apply the same measure to summaries with different combinations and expressions of identifying cues.

Answer to RQ1: Under the fixed evaluation protocol, $N(D)$ provides a common quantitative measure of re-identification risk for correctly resolved mobility summaries, while terminal-state diagnostics represent cases in which $N(D)$ is undefined.

B. Effects of Numerical Perturbation (RQ2)

Figure 2 visualizes the terminal-state counts across the seven numerical-perturbation conditions.

In Experiment 1, the evaluator correctly resolved 99 of 100 targets under No Perturbation, $\epsilon = 100$, and $\epsilon = 10$, 96 under $\epsilon = 1$, 72 under $\epsilon = 0.1$, 31 under $\epsilon = 0.01$, and 6 under $\epsilon = 0.001$. False isolation occurred in 1, 1, 1, 3, 28, 65, and 82 runs, respectively. Zero-information-gain termination occurred in 0, 0, 0, 1, 0, 4, and 12 runs, respectively. Max-step termination occurred in 0 runs for all seven conditions. Mean $N(D)$ among correctly resolved runs was 11.63, 11.58, 11.68, 11.54, 11.97, 13.13, and 17.83 (the last value based on only six correctly resolved targets); mean questions over all targets was 11.65, 11.60, 11.70, 11.63, 12.12, 14.03, and 15.56. Mean $N(D)$ and mean question counts did not change monotonically

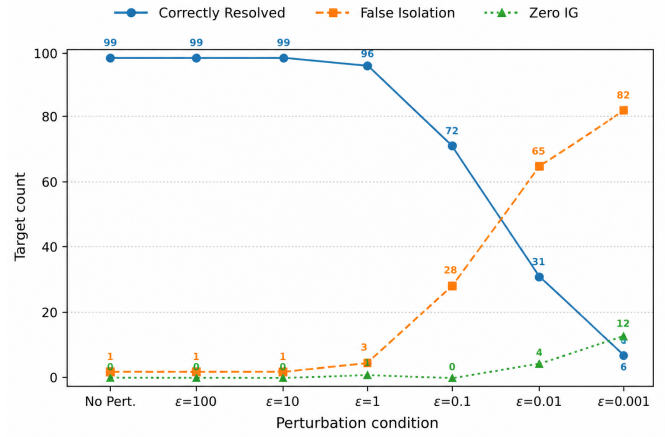


Fig. 2. Terminal-state counts across numerical-perturbation conditions. Correct resolution generally decreases under stronger tested perturbations, while false isolation increases substantially. Zero-information-gain termination appears primarily under the strongest tested perturbation levels.

TABLE I
NUMERICAL PERTURBATION RESULTS.

Condition	Resolved	False Iso.	Zero IG	Max Step	Mean $N(D)^{\dagger}$	Mean Qs. [‡]
No Perturbation	99	1	0	0	11.63	11.65
$\epsilon = 100$	99	1	0	0	11.58	11.60
$\epsilon = 10$	99	1	0	0	11.68	11.70
$\epsilon = 1$	96	3	1	0	11.54	11.63
$\epsilon = 0.1$	72	28	0	0	11.97	12.12
$\epsilon = 0.01$	31	65	4	0	13.13	14.03
$\epsilon = 0.001$	6	82	12	0	17.83	15.56

$N(D)$, resolved only, computed over correctly resolved targets. [‡]Mean Questions, all runs, computed over all 100 runs. Target elimination and Oracle-related disagreement counts are reported separately in Section VI-D because they may co-occur with the terminal outcomes shown here.

across all perturbation levels. In contrast, correct resolution was non-increasing and false isolation was non-decreasing as the tested perturbation became stronger.

For the target-aligned No Perturbation vs. $\epsilon = 1$ comparison, question counts decreased for 19 targets, increased for 14, and were unchanged for 67, with answer paths diverging for 37 targets; these opposing target-level changes nearly canceled in the aggregate means.

Answer to RQ2: At the strongest tested perturbation level, the mean $N(D)$ among correctly resolved runs increased from 11.63 to 17.83, while the correct-identification rate decreased from 99% to 6%. Thus, the strongest perturbation substantially reduced overall identification success and increased the conditional search cost among the few correctly resolved runs. However, $N(D)$ did not change monotonically across all perturbation levels.

C. Effects of Attribute Removal and Prompt Abstraction (RQ3)

a) *Prompt abstraction:* The prompt-granularity evaluation examines the effect of output abstraction. Table II reports preliminary diagnostic results for the 100-target prompt-

TABLE II
PROMPT-GRANULARITY RESULTS

Level	Correctly Resolved	False Isolation	Zero IG	Max-step	Mean $N(D)$	Mean Unknown Steps
Level 1	82	18	0	0	14.96	7.62
Level 2	71	29	0	0	26.17	19.76
Level 3	7	17	43	33	41.43	114.49

TABLE III
ATTRIBUTE-ABLATION RESULTS

Condition	Correctly Resolved	False Isolation	Zero IG	Max-step	Mean $N(D)$	Mean Unknown Steps
Full Baseline	76	24	0	0	15.20	8.95
w/o Location	42	30	28	0	20.64	19.09
w/o Time	66	3	31	0	40.71	38.88
w/o Frequency	77	20	3	0	35.64	29.55

granularity experiment (Level 1 Detailed, Level 2 Moderate, Level 3 Abstract) with 200 candidates and a 200-question budget. From Level 1 to Level 3, correct resolution fell from 82 to 7, Mean $N(D)$ rose from 14.96 to 41.43, and Mean Unknown Steps rose from 7.62 to 114.49. At Level 3, the remaining outcomes were 17 false isolations, 43 zero-information-gain terminations, and 33 maximum-step terminations.

The four tabled outcomes form an exclusive partition, while target elimination is a separate, potentially co-occurring diagnostic flag. Because Level 3 Mean $N(D)$ uses only seven resolved runs, it must be interpreted with the unresolved outcomes.

b) Attribute removal: Table III compares the Full Baseline with three attribute-ablation settings. The Full Baseline correctly resolved 76 of 100 targets. Removing location reduced the number of correctly resolved targets to 42, whereas removing time and frequency resulted in 66 and 77 correctly resolved targets, respectively. The resolved-only mean $N(D)$ values were 15.20, 20.64, 40.71, and 35.64. Zero-information-gain termination occurred 0, 28, 31, and 3 times, respectively. The corresponding mean unknown steps over all targets were 8.95, 19.09, 38.88, and 29.55. Max-step termination occurred in 0 runs for all four conditions.

Answer to RQ3: Attribute removal and prompt abstraction changed identification rates, question costs, and failure modes, with different effects across attributes and abstraction levels.

D. Interpreting Risk under Failure Modes (RQ4)

Figure 2 and Table I summarize the terminal outcomes for Experiment 1, while Tables II and III report the corresponding outcomes for prompt abstraction and attribute removal. Experiment 1 includes correct resolution, false isolation, and, for $\epsilon = 1$, $\epsilon = 0.01$, and $\epsilon = 0.001$, zero-information-gain termination. The prompt-granularity experiment exhibits a wider range of terminal states: at Level 3, Table II shows 7 correctly resolved runs, 17 false isolations, 43 zero-information-gain terminations, and 33 terminations at the question limit. Target elimination is not a tabled terminal state; it is recorded separately as a diagnostic flag because it can co-occur with these terminal-state labels. Across the seven Experiment 1

conditions, target elimination occurred in 1, 1, 1, 4, 28, 68, and 94 runs, while Mean Unknown Steps over all 100 targets was 3.97, 3.90, 4.02, 3.98, 4.43, 6.47, and 8.08, respectively. The corresponding total Oracle-error counts were 1, 1, 1, 4, 32, 102, and 246; no run ended with an empty candidate set.

Trace inspection also revealed Oracle-dependent error modes: in some location questions, the summary explicitly stated a different value for the same feature, yet the Oracle returned `unknown` instead of `no`; in other cases, an incorrect Oracle answer removed the true target. Such cases are recorded as false isolation or target elimination rather than as successful identifications.

The completed runs show that a sole remaining candidate does not ensure correct re-identification. We therefore verify whether that candidate is the target and distinguish correct identification from false isolation, target elimination, and evidence exhaustion; only correct identification is counted as a successful outcome.

Because Mean $N(D)$ is conditional on resolution, the value 17.83 in Table I is based on only six resolved runs. Similarly, the value 41.43 at prompt Level 3 in Table II is based on seven resolved runs, while 93 runs were unsuccessful. These conditional means therefore require the correct-resolution counts and terminal outcomes for interpretation.

For deployment, $N(D)$ can flag summaries with low anonymity cost, but diagnostics are needed to distinguish genuine audit resistance from high conditional means shaped by budget exhaustion, false isolation, or target elimination.

Answer to RQ4: Failure modes can make a high mean $N(D)$ among correctly resolved runs appear to indicate stronger privacy because unsuccessful runs are excluded from the calculation, changing the composition of the resolved subset. Therefore, $N(D)$ must be interpreted together with the correct-resolution rate and failure diagnostics, including false isolation, target elimination, and evidence exhaustion.

E. Integrated Discussion

Taken together, RQ1–RQ4 show that the proposed method resolves the central evaluation challenge within the study setting. $N(D)$ converts heterogeneous identifying cues into a common question-based cost, allowing mobility summaries produced under numerical perturbation, attribute removal, and prompt abstraction to be assessed under the same protocol. Terminal-state diagnostics complement this cost by distinguishing correct identification from false isolation, target elimination, and evidence exhaustion. The method therefore supports consistent assessment across heterogeneous summaries and privacy transformations while accounting for both successful and unsuccessful identification outcomes.

F. Limitations

This study has several limitations. The evaluation uses one month of Android data from a single region and a fixed candidate pool, which limits generalizability. The post-aggregation Laplace perturbation of aggregated count and frequency attributes is an experimental baseline rather than

a formal differential-privacy or geo-indistinguishability guarantee. The results also depend on a single LLM-based Oracle, and Mean $N(D)$ is conditional on correct resolution and must therefore be interpreted with resolution and failure diagnostics. The prompt-granularity results remain preliminary because this run was executed separately from the main experiment and its correspondence to a later prompt-fix revision could not be independently established; these preliminary results are excluded from the main reproducibility claim. The attribute-ablation experiment is also exploratory, and $N(D)$ has not yet been compared with existing privacy metrics or idealized identification baselines. In addition, the information-gain selection rule is deterministic only given fixed Oracle outputs, while the end-to-end audit may vary with LLM inference behavior. The recorded target-profile disagreement counts should not be interpreted as a direct estimate of Oracle accuracy, particularly under numerical perturbation.

VII. CONCLUSION

Even without direct identifiers, analytical reports may reveal information that enables individual re-identification. We therefore proposed $N(D)$ —the number of iterative questions required for correct identification—together with terminal-state diagnostics to quantify this risk in LLM-generated mobility summaries.

We evaluated 100 targets against a fixed 200-candidate pool under post-aggregation numerical perturbation of aggregated count and frequency attributes, attribute removal, and prompt abstraction, with a 200-question limit. At the strongest perturbation ($\epsilon = 0.001$), correct identification fell from 99% to 6% and false isolation rose from 1% to 82%, while mean $N(D)$ among correctly resolved runs increased from 11.63 to 17.83, based on six correctly resolved runs in the strongest condition. Attribute removal and prompt abstraction also changed identification rates, question costs, and failure modes.

Together, these results show that the method resolves the central evaluation challenge within the study setting. $N(D)$ maps heterogeneous identifying cues to a common cost, while terminal-state diagnostics distinguish correct identification from false isolation, target elimination, and evidence exhaustion. The method therefore assesses heterogeneous summaries and privacy transformations while accounting for both successful and unsuccessful outcomes. In particular, $N(D)$ quantifies the re-identification risk that persists after names and user IDs are removed from analytical reports, as demonstrated here for LLM-generated mobility summaries.

Future work will test generalization beyond mobility summaries and fixed candidate pools, develop adaptive questions, assess robustness across Oracle models and prompting strategies, and compare $N(D)$ with established privacy metrics and idealized identification baselines.

ACKNOWLEDGMENT

The authors thank the Research Center for Advanced Technology and GX and Yasuhiro Mimura for providing data

access and valuable advice. This work was supported by JSPS KAKENHI Grant Number 24K05080. As part of the experimental protocol described in Section V-A, Google’s Gemini-3.1-flash-lite was used as the summary generator and summary-based Oracle. OpenAI’s ChatGPT (GPT-5.5) supported language refinement during manuscript preparation. The authors reviewed all generated outputs and take full responsibility for the analyses, writing, and manuscript content.

REFERENCES

- [1] G. Heiler, T. Reisch, J. Hurt, M. Forghani, A. Omani, A. Hanbury, and F. Karimipour, “Country-wide mobility changes observed using mobile phone data during COVID-19 pandemic,” in *2020 IEEE International Conference on Big Data (Big Data)*, 2020, pp. 3123–3132.
- [2] Y. Li, D. Yang, and X. Hu, “A differential privacy-based privacy-preserving data publishing algorithm for transit smart card data,” *Transportation Research Part C: Emerging Technologies*, vol. 115, p. 102634, 2020.
- [3] F. Doshi-Velez and B. Kim, “Towards a rigorous science of interpretable machine learning,” *arXiv preprint arXiv:1702.08608*, 2017.
- [4] A. D. Palù, A. Duvier, and A. Formisano, “Towards explainable Data-to-Text generation,” in *CEUR Workshop Proceedings*, vol. 3428. CEUR-WS, 2023.
- [5] A. Gatt and E. Krahmer, “Survey of the state of the art in natural language generation: Core tasks, applications and evaluation,” *Journal of Artificial Intelligence Research*, vol. 61, pp. 65–170, 2018.
- [6] OpenAI, “GPT-4 technical report,” *arXiv preprint arXiv:2303.08774*, 2023.
- [7] T. Aksu, C. Liu, A. Saha, S. Tan, C. Xiong, and D. Sahoo, “XForecast: Evaluating natural language explanations for time series forecasting,” *arXiv preprint arXiv:2410.14180*, 2024.
- [8] P. Lison, I. Pilán, D. Sanchez, M. Batet, and L. Øvrelid, “Anonymisation models for text data: State of the art, challenges and future directions,” in *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*. Association for Computational Linguistics, 2021, pp. 4188–4203.
- [9] Y.-A. de Montjoye, C. A. Hidalgo, M. Verleysen, and V. D. Blondel, “Unique in the crowd: The privacy bounds of human mobility,” *Scientific Reports*, vol. 3, p. 1376, 2013.
- [10] C. Dwork, “Differential privacy,” in *Automata, Languages and Programming*. Springer Berlin Heidelberg, 2006, pp. 1–12.
- [11] M. E. Andrés, N. E. Bordenabe, K. Chatzikokolakis, and C. Palamidessi, “Geo-indistinguishability: Differential privacy for location-based systems,” in *Proceedings of the 2013 ACM SIGSAC Conference on Computer and Communications Security*. Association for Computing Machinery, 2013, pp. 901–914.
- [12] I. Pilán, P. Lison, L. Øvrelid, A. Papadopoulou, D. Sánchez, and M. Batet, “The text anonymization benchmark (TAB): A dedicated corpus and evaluation framework for text anonymization,” *Computational Linguistics*, vol. 48, no. 4, pp. 1053–1101, 2022.
- [13] A. Hughes, N. Aletras, and N. Ma, “How private are language models in abstractive summarization?” in *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, 2025, pp. 30 112–30 130.
- [14] L. G. G. Charpentier and P. Lison, “Re-identification of de-identified documents with autoregressive infilling,” in *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Association for Computational Linguistics, 2025, pp. 1192–1209.
- [15] M. De Bruyn, E. Lotfi, J. Buhmann, and W. Daelemans, “Is it smaller than a tennis ball? language models play the game of twenty questions,” in *Proceedings of the Fifth BlackboxNLP Workshop on Analyzing and Interpreting Neural Networks for NLP*. Association for Computational Linguistics, 2022, pp. 80–90.
- [16] R. D. Nowak, “The geometry of generalized binary search,” *IEEE Transactions on Information Theory*, vol. 57, no. 12, pp. 7893–7906, 2011.